

AWS first challenge

- Launch one EC2 instance using Amazon Linux 2.
- Install Docker.
- Install Jenkins.
- Install Apache.
- Install Nginx.
- Install Apache Tomcat.

Everything needs to be setup and running on the **same EC2**.

Objective:

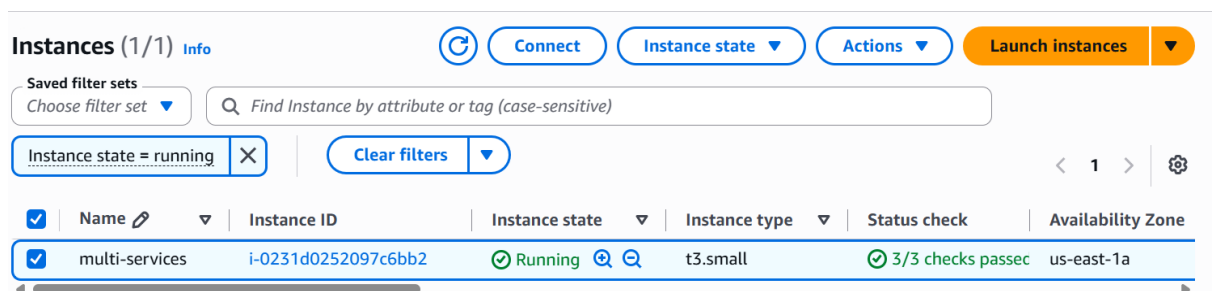
Launch one Amazon EC2 instance using **Amazon Linux 2** and install **Docker, Jenkins, Apache, Nginx, and Apache Tomcat** on the same server.

Purpose:

To understand how multiple services can be installed, configured, and run simultaneously on a single EC2 instance using different port numbers.

Implementation:

1. Launched an **Amazon Linux 2 EC2 instance**.



Instances (1/1) Info							Connect Instance state Actions Launch instances	
Saved filter sets		Find Instance by attribute or tag (case-sensitive)						
Choose filter set								
Instance state = running		Clear filters						
☒	Name ✎	Instance ID	Instance state	Instance type	Status check	Availability Zone		
☒	multi-services	i-0231d0252097c6bb2	Running	t3.small	3/3 checks passed	us-east-1a		

2. Connected to the EC2 instance using SSH.


```
[root@ip-172-31-10-62 ~]# exit
logout
[ec2-user@ip-172-31-10-62 ~]$ exit
logout
Connection to 3.234.182.127 closed.

91738@IDEAPAD MINGW64 ~/Downloads
$ ssh -i linux-testing.pem ec2-user@3.234.182.127

#_
~\##### Amazon Linux 2023
~~\#####\
~~\###|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~'~'~>
~~~
~~~.~.~
~~~/~'~>
~/_/~/~'~>

Last login: Mon Aug 24 09:26:35 2026 from 103.203.172.230
[ec2-user@ip-172-31-10-62 ~]$ sudo su -
Last login: Mon Aug 24 09:26:44 UTC 2026 on pts/1
[root@ip-172-31-10-62 ~]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-10-62 ~]# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS          NAMES
[root@ip-172-31-10-62 ~]#
```

```
[root@ip-172-31-10-62 ~]# java -version
openjdk version "21.0.12" 2026-07-21 LTS
OpenJDK Runtime Environment Corretto-21.0.12.8.1 (build 21.0.12+8-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.12.8.1 (build 21.0.12+8-LTS, mixed mode, sharing)
[root@ip-172-31-10-62 ~]# wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/rpm-stable/jenkins.repo
2026-08-24 09:43:57 -- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.38.133, 2a04:4e42:79::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)[146.75.38.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'

/etc/yum.repos.d/jenkins.repo 100%[=====] 267 --.-KB/s in 0s

2026-08-24 09:43:57 (21.3 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]

[root@ip-172-31-10-62 ~]# sudo rpm --import https://pkg.jenkins.io/rpm-stable/jenkins.io-2026.key
[root@ip-172-31-10-62 ~]# yum upgrade -y
jenkins-stable
jenkins-stable 11 kB/s | 833 B 00:00
100 kB/s | 1.6 kB 00:00
Importing GPG key 0x14ABFC68:
Userid : "jenkins Project <jenkinsci-board@googlegroups.com>"
Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 719F F487 14AB FC68
From : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
jenkins-stable 31 kB/s | 2.6 kB 00:00
Dependencies resolved.
Nothing to do.
Complete!
```

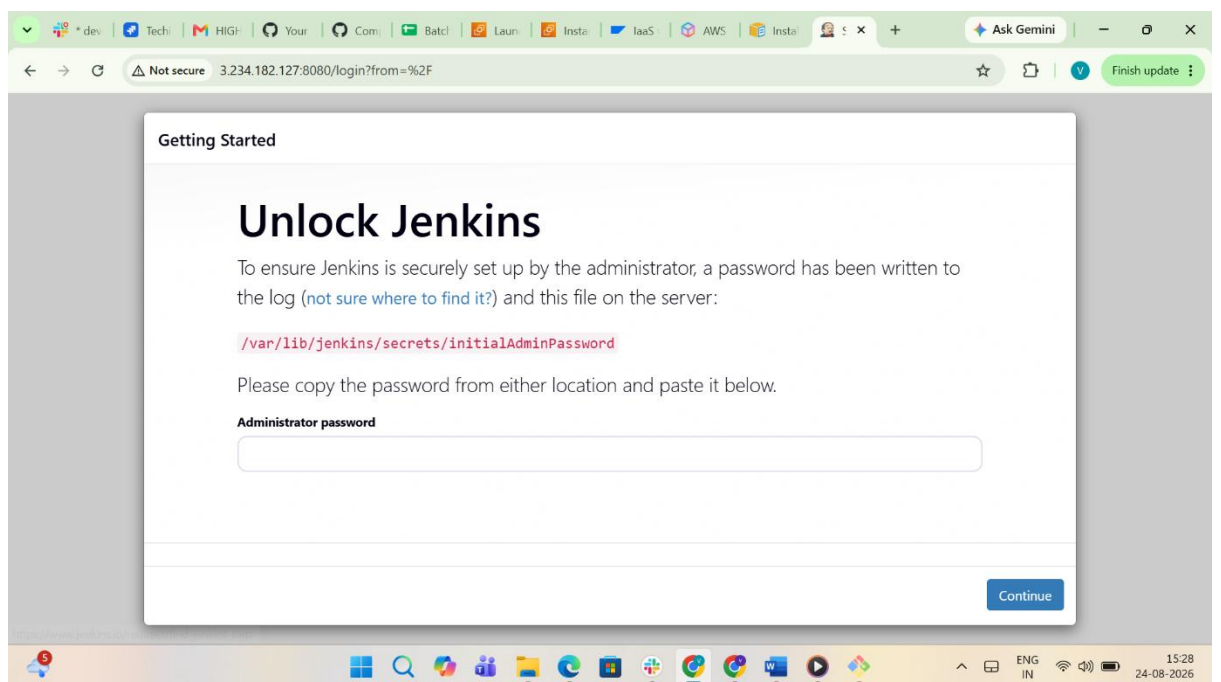
```
[root@ip-172-31-10-62 ~]# sudo rpm --import https://pkg.jenkins.io/rpm-stable/jenkins.io-2026.key
[root@ip-172-31-10-62 ~]# yum upgrade -y
jenkins-stable                               11 kB/s | 833 B    00:00
jenkins-stable                               100 kB/s | 1.6 kB  00:00
Importing GPG key 0x14ABFC68:
  Userid : "Jenkins Project <jenkinsci-board@googlegroups.com>"
  Fingerprint: SE38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
  From : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
jenkins-stable                               31 kB/s | 2.6 kB  00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-10-62 ~]# yum install jenkins -y
Last metadata expiration check: 0:00:39 ago on Mon Aug 24 09:44:37 2026.
Dependencies resolved.

Package                Architecture      Version      Repository      Size
Installing:
jenkins                noarch            2.568.2-1   jenkins         96 M
Transaction Summary
Install 1 Package
Total download size: 96 M
Installed size: 96 M
Downloading Packages:
jenkins-2.568.2-1.noarch.rpm                77 MB/s | 96 MB    00:01
Total
77 MB/s | 96 MB    00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
```

```
Installed:
jenkins-2.568.2-1.noarch

Complete!
[root@ip-172-31-10-62 ~]# systemctl enable jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
[root@ip-172-31-10-62 ~]# systemctl start jenkins
[root@ip-172-31-10-62 ~]# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-08-24 09:47:10 UTC; 25s ago
     Main PID: 29147 (java)
       Tasks: 48 (limit: 2255)
        Memory: 500.3M
          CPU: 23.663s
     CGroup: /system.slice/jenkins.service
             └─29147 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Aug 24 09:47:04 ip-172-31-10-62.ec2.internal jenkins[29147]: [LF]>
Aug 24 09:47:04 ip-172-31-10-62.ec2.internal jenkins[29147]: [LF]> *****
Aug 24 09:47:04 ip-172-31-10-62.ec2.internal jenkins[29147]: [LF]> *****
Aug 24 09:47:10 ip-172-31-10-62.ec2.internal jenkins[29147]: 2026-08-24 09:47:10.088+0000 [id=40] INFO jenkins.InitReactorRunner$1#onAttained: Completed init
Aug 24 09:47:10 ip-172-31-10-62.ec2.internal jenkins[29147]: 2026-08-24 09:47:10.113+0000 [id=30] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully u
Aug 24 09:47:10 ip-172-31-10-62.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Aug 24 09:47:10 ip-172-31-10-62.ec2.internal jenkins[29147]: 2026-08-24 09:47:10.320+0000 [id=56] INFO h.m.DownloadService$Downloadable#load: Obtained the up
Aug 24 09:47:10 ip-172-31-10-62.ec2.internal jenkins[29147]: 2026-08-24 09:47:10.321+0000 [id=56] INFO hudson.util.Retrier#start: Performed the action check
Aug 24 09:47:15 ip-172-31-10-62.ec2.internal jenkins[29147]: 2026-08-24 09:47:15.171+0000 [id=75] WARNING h.n.DiskSpaceMonitorDescriptor#markNodeOfflineOrOnl
lines 1-20/20 (END)
```



6. Installed and configured **Apache HTTP Server**.

```
root@ip-172-31-10-62:~  
[root@ip-172-31-10-62 ~]# yum install httpd  
Last metadata expiration check: 0:16:58 ago on Mon Aug 24 09:44:37 2026.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
httpd	x86_64	2.4.68-1.amzn2023.0.1	amazonlinux	46 k
Installing dependencies:				
apr	x86_64	1.7.5-1.amzn2023.0.4	amazonlinux	129 k
apr-util	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	97 k
apr-util-ldap	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	13 k
generic-logos-httpd	noarch	18.0.0-12.amzn2023.0.3	amazonlinux	19 k
httpd-core	x86_64	2.4.68-1.amzn2023.0.1	amazonlinux	1.4 M
httpd-filesystem	noarch	2.4.68-1.amzn2023.0.1	amazonlinux	12 k
httpd-tools	x86_64	2.4.68-1.amzn2023.0.1	amazonlinux	80 k
mailcap	noarch	2.1.49-3.amzn2023.0.3	amazonlinux	33 k
Installing weak dependencies:				
apr-util-openssl	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	15 k
mod_http2	x86_64	2.0.42-1.amzn2023.0.1	amazonlinux	167 k
mod_lua	x86_64	2.4.68-1.amzn2023.0.1	amazonlinux	59 k

```
=====
```

Transaction Summary			
Install	12 Packages		
Total download size: 2.0 M			
Installed size: 6.2 M			
Is this ok [y/N]: yes			
Downloading Packages:			
(1/12): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm	3.0 MB/s 129 kB	00:00	
(2/12): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm	289 kB/s 13 kB	00:00	
(3/12): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm	1.9 MB/s 97 kB	00:00	
(4/12): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm	596 kB/s 15 kB	00:00	
(5/12): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm	694 kB/s 19 kB	00:00	
(6/12): httpd-2.4.68-1.amzn2023.0.1.x86_64.rpm	1.7 MB/s 46 kB	00:00	
(7/12): httpd-core-2.4.68-1.amzn2023.0.1.x86_64.rpm	34 MB/s 1.4 MB	00:00	
(8/12): httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch.rpm	319 kB/s 12 kB	00:00	
(9/12): httpd-tools-2.4.68-1.amzn2023.0.1.x86_64.rpm	2.1 MB/s 80 kB	00:00	
(10/12): mod_http2-2.0.42-1.amzn2023.0.1.x86_64.rpm	6.4 MB/s 167 kB	00:00	
(11/12): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm	1.1 MB/s 33 kB	00:00	
(12/12): mod_lua-2.4.68-1.amzn2023.0.1.x86_64.rpm	2.0 MB/s 59 kB	00:00	

Total	11 MB/s 2.0 MB	00:00	

```
=====
```

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.

```
Installed:  
apr-1.7.5-1.amzn2023.0.4.x86_64          apr-util-1.6.3-1.amzn2023.0.2.x86_64          apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64  
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  httpd-2.4.68-1.amzn2023.0.1.x86_64  
httpd-core-2.4.68-1.amzn2023.0.1.x86_64      httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch      httpd-tools-2.4.68-1.amzn2023.0.1.x86_64  
mailcap-2.1.49-3.amzn2023.0.3.noarch          mod_http2-2.0.42-1.amzn2023.0.1.x86_64          mod_lua-2.4.68-1.amzn2023.0.1.x86_64  
  
Complete!  
[root@ip-172-31-10-62 ~]# systemctl enable httpd  
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service - /usr/lib/systemd/system/httpd.service.  
[root@ip-172-31-10-62 ~]# systemctl start httpd  
[root@ip-172-31-10-62 ~]# systemctl status httpd  
● httpd.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)  
   Active: active (running) since Mon 2026-08-24 10:02:29 UTC; 29s ago  
     Docs: man:httpd.service(8)  
  Main PID: 30153 (httpd)  
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:  0 B/sec"  
    Tasks: 177 (limit: 2255)  
   Memory: 14.0M  
      CPU: 85ms  
   CGroup: /system.slice/httpd.service  
           └─30153 /usr/sbin/httpd -DFOREGROUND  
           └─30154 /usr/sbin/httpd -DFOREGROUND  
           └─30155 /usr/sbin/httpd -DFOREGROUND  
           └─30157 /usr/sbin/httpd -DFOREGROUND  
           └─30177 /usr/sbin/httpd -DFOREGROUND  
  
Aug 24 10:02:29 ip-172-31-10-62.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...  
Aug 24 10:02:29 ip-172-31-10-62.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.  
Aug 24 10:02:29 ip-172-31-10-62.ec2.internal httpd[30153]: Server configured, listening on: port 80  
[root@ip-172-31-10-62 ~]# systemctl is-active httpd  
active
```

```
✓ *de | Tech | HIG | You | Con | Bat | Laur | Inst | laaS | AW | Inst | Sigr | x | + | Ask Gemini | - | x  
← → ↻ ⚠ Not secure 3.234.182.127 ☆ 📄 🔍 Finish update
```

It works!



7. Installed and configured Nginx.

```
root@ip-172-31-10-62:~  
[root@ip-172-31-10-62 ~]# yum install nginx -y  
Last metadata expiration check: 0:20:45 ago on Mon Aug 24 09:44:37 2026.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing: nginx	x86_64	1:1.30.4-1.amzn2023.0.1	amazonlinux	34 k
Installing dependencies:				
gperftools-libs	x86_64	2.9.1-1.amzn2023.0.3	amazonlinux	308 k
libunwind	x86_64	1.4.0-5.amzn2023.0.3	amazonlinux	66 k
nginx-core	x86_64	1:1.30.4-1.amzn2023.0.1	amazonlinux	711 k
nginxfilesystem	noarch	1:1.30.4-1.amzn2023.0.1	amazonlinux	10 k
nginx-mimetypes	noarch	2.1.49-3.amzn2023.0.3	amazonlinux	21 k

```
=====
```

Transaction Summary

Install 6 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Downloading Packages:

(1/6): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm				
(2/6): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm	7.6 MB/s	308 kB	00:00	
(3/6): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm	792 kB/s	34 kB	00:00	
(4/6): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm	1.4 MB/s	66 kB	00:00	
(5/6): nginxfilesystem-1.30.4-1.amzn2023.0.1.noarch.rpm	23 MB/s	711 kB	00:00	
(6/6): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm	354 kB/s	10 kB	00:00	
	730 kB/s	21 kB	00:00	

Total
9.5 MB/s | 1.1 MB | 00:00

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Running scriptlet: nginxfilesystem-1:1.30.4-1.amzn2023.0.1.noarch 1/1
Installing : nginxfilesystem-1:1.30.4-1.amzn2023.0.1.noarch 1/6
Installing : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch 2/6
Installing : libunwind-1.4.0-5.amzn2023.0.3.x86_64 3/6
Installing : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 4/6
Installing : nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64 5/6
Installing : nginx-1:1.30.4-1.amzn2023.0.1.x86_64 6/6
Running scriptlet: nginx-1:1.30.4-1.amzn2023.0.1.x86_64 6/6
Verifying : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 1/6
Verifying : libunwind-1.4.0-5.amzn2023.0.3.x86_64 2/6
Verifying : nginx-1:1.30.4-1.amzn2023.0.1.x86_64 3/6
Verifying : nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64 4/6

```
root@ip-172-31-10-62:~  
Installing: gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 libunwind-1.4.0-5.amzn2023.0.3.x86_64 nginx-1:1.30.4-1.amzn2023.0.1.x86_64  
nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64 nginxfilesystem-1:1.30.4-1.amzn2023.0.1.noarch nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch  
Complete!  
[root@ip-172-31-10-62 ~]# cat /etc/nginx/nginx.conf  
# For more information on configuration, see:  
# * Official English Documentation: http://nginx.org/en/docs/  
# * Official Russian Documentation: http://nginx.org/ru/docs/  
  
user nginx;  
worker_processes auto;  
error_log /var/log/nginx/error.log notice;  
pid /run/nginx.pid;  
  
# Load dynamic modules. See /usr/share/doc/nginx/README.dynamic.  
include /usr/share/nginx/modules/*.conf;  
  
events {  
    worker_connections 1024;  
}  
  
http {  
    log_format main '$remote_addr - $remote_user [$time_local] "$request" '  
        '$status $body_bytes_sent "$http_referer" '  
        '"$http_user_agent" "$http_x_forwarded_for"';  
  
    access_log /var/log/nginx/access.log main;  
  
    sendfile on;  
    tcp_nopush on;  
    keepalive_timeout 65;  
    types_hash_max_size 4096;  
  
    include /etc/nginx/mime.types;  
    default_type application/octet-stream;  
  
    # Load modular configuration files from the /etc/nginx/conf.d directory.  
    # See http://nginx.org/en/docs/nginx_core_module.html#include  
    # for more information.  
    include /etc/nginx/conf.d/*.conf;  
  
    server {  
        listen 80;  
        listen [::]:80;  
        server_name _;  
        root /usr/share/nginx/html;  
    }  
}
```

```
[root@ip-172-31-10-62 ~]# nano /etc/nginx/nginx.conf  
[root@ip-172-31-10-62 ~]# systemctl daemon-reload  
[root@ip-172-31-10-62 ~]# systemctl stop nginx  
[root@ip-172-31-10-62 ~]# systemctl restart nginx  
[root@ip-172-31-10-62 ~]# systemctl status nginx  
● nginx.service - The nginx HTTP and reverse proxy server  
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)  
   Active: active (running) since Mon 2026-08-24 10:18:38 UTC; 27s ago  
     Process: 31867 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)  
     Process: 31868 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)  
     Process: 31869 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)  
    Main PID: 31870 (nginx)  
       Tasks: 3 (limit: 2255)  
      Memory: 3.4M  
         CPU: 65ms  
      CGroup: /system.slice/nginx.service  
              └─31870 "nginx: master process /usr/sbin/nginx"  
                └─31871 "nginx: worker process"  
                  └─31872 "nginx: worker process"  
  
Aug 24 10:18:38 ip-172-31-10-62.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...  
Aug 24 10:18:38 ip-172-31-10-62.ec2.internal nginx[31868]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
Aug 24 10:18:38 ip-172-31-10-62.ec2.internal nginx[31868]: nginx: configuration file /etc/nginx/nginx.conf test is successful  
Aug 24 10:18:38 ip-172-31-10-62.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
```



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.

To engage with the community please visit community.nginx.org.

For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

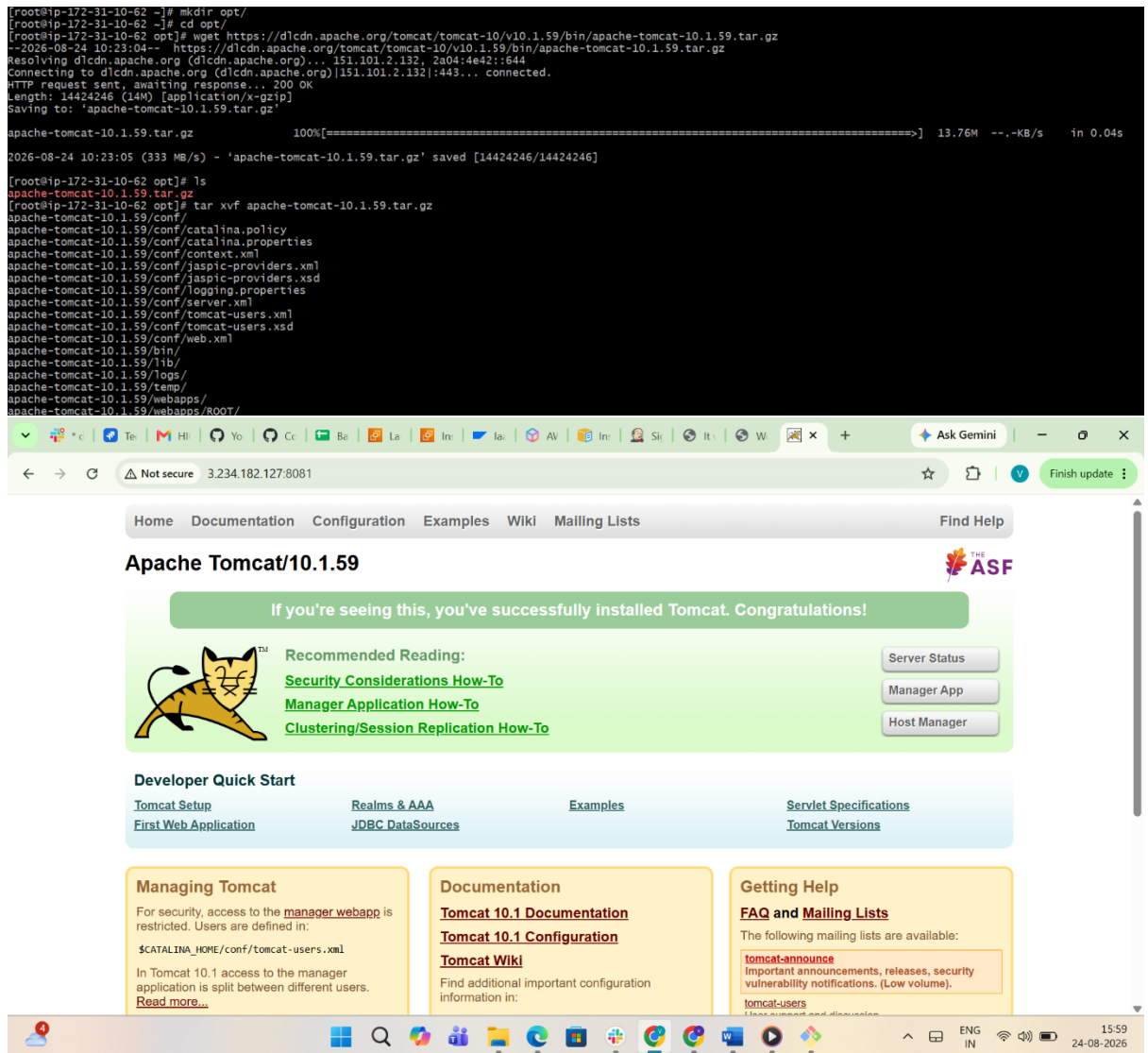


8. Installed and configured Apache Tomcat.

```
[root@ip-172-31-10-62 opt]# mv apache-tomcat-10.1.59 tomcat
[root@ip-172-31-10-62 opt]# ls
apache-tomcat-10.1.59.tar.gz tomcat
[root@ip-172-31-10-62 opt]# cd tomcat
[root@ip-172-31-10-62 tomcat]# ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
[root@ip-172-31-10-62 tomcat]# cd conf
[root@ip-172-31-10-62 conf]# ls
catalina.policy catalina.properties context.xml jaspic-providers.xml jaspic-providers.xsd logging.properties server.xml tomcat-users.xml tomcat-users.xsd web.xml
[root@ip-172-31-10-62 conf]# cat server.xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
Licensed to the Apache Software Foundation (ASF) under one or more
contributor license agreements.  See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License.  You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<!-- Note: A "Server" is not itself a "Container", so you may not
define subcomponents such as "Valves" at this level.
Documentation at /docs/config/server.html
-->
<Server port="8005" shutdown="SHUTDOWN">
  <Listener className="org.apache.catalina.startup.VersionLoggerListener" />
  <!-- Security listener. Documentation at /docs/config/listeners.html
  <Listener className="org.apache.catalina.security.SecurityListener" />
  -->
  <!-- OpenSSL support using Tomcat Native -->
  <Listener className="org.apache.catalina.core.AprLifecycleListener" />
  <!-- OpenSSL support using FFmpeg API from Java 22 -->
  <!-- <Listener className="org.apache.catalina.core.OpenSSLLifecycleListener" /> -->
  <!-- Prevent memory leaks due to use of particular java/javax APIs-->
```



9. Assigned different ports to avoid port conflicts.

Example:

Service	Port
Apache	80
Nginx	81
Jenkins	8080
Tomcat	8081

10. Configured the required **EC2 Security Group** inbound rules for the service ports.

Inbound rules	Outbound rules	Sharing	VPC associations	Related resources - new	Tags
---------------	----------------	---------	------------------	-------------------------	------

Inbound rules (5)							Manage tags	Edit inbound rules
Search								
<	1	>						
Security group rule ID	IP version	Type	Protocol	Port range				
sgr-033fc7819d6e5fcaf	IPv4	Custom TCP	TCP	8081				
sgr-05ec40745afe95173	IPv4	Custom TCP	TCP	8080				
sgr-00b7438927d08aaed	IPv4	SSH	TCP	22				
sgr-0668050825beaae54	IPv4	HTTP	TCP	80				
sgr-0f31594a119ce8fd5	IPv4	Custom TCP	TCP	81				

11. Verified the running services using:

systemctl status httpd

systemctl status nginx

systemctl status jenkins

systemctl status tomcat

For Docker:

systemctl status docker

```
[root@ip-172-31-10-62 tomcat]# systemctl is-active tomcat
active
[root@ip-172-31-10-62 tomcat]# systemctl is-active nginx
active
[root@ip-172-31-10-62 tomcat]# systemctl is-active httpd
active
[root@ip-172-31-10-62 tomcat]# systemctl is-active jenkins
active
[root@ip-172-31-10-62 tomcat]# systemctl is-active docker
active
```

12. Tested each service using the EC2 public IP and configured port:

http://<EC2-PUBLIC-IP>:80

http://<EC2-PUBLIC-IP>:81

http://<EC2-PUBLIC-IP>:8080

http://<EC2-PUBLIC-IP>:8081

Conclusion:

Successfully launched a single **Amazon Linux 2 EC2 instance** and configured **Docker, Jenkins, Apache, Nginx, and Apache Tomcat** to run simultaneously using separate ports.